Differential Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$.

Separate: $\frac{dy}{y} = \frac{dx}{x}$. Integrate: $\ln|y| = \ln|x| + C$, so $y = Ax$. Using $y = 6, x = 2$: $A = 3$. Solution: $y = 3x$.

2 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = 3x^2 y$, given $y = 2$ when $x = 0$.

Separate: $\frac{dy}{y} = 3x^2\,dx$. Integrate: $\ln|y| = x^3 + C$, so $y = Ae^{x^3}$. Using $y = 2, x = 0$: $A = 2$. Solution: $y = 2e^{x^3}$.

3 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = \frac{2x}{y^2}$, given $y = 2$ when $x = 0$.

Separate: $y^2\,dy = 2x\,dx$. Integrate: $\frac{y^3}{3} = x^2 + C$. Using $y = 2, x = 0$: $C = \frac{8}{3}$. So $y^3 = 3x^2 + 8$, giving $y = (3x^2 + 8)^{1/3}$.

4 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$, given $y = 1$ when $x = 1$.

Separate: $\frac{dy}{y^2} = \frac{dx}{x}$. Integrate: $-\frac{1}{y} = \ln|x| + C$. Using $y = 1, x = 1$: $C = -1$. So $-\frac{1}{y} = \ln x - 1$, giving $y = \frac{1}{1 - \ln x}$.

5 **[Foundational]** • **Target: Forming differential equations from a description**

The rate of decrease of a quantity Q is proportional to Q itself. Write a differential equation for $\frac{dQ}{dt}$, using $k > 0$ as the constant of proportionality.

$\frac{dQ}{dt} = -kQ$.

6 **[Foundational]** • **Target: Forming differential equations from a description**

A tank drains so that the rate of decrease of its volume V is proportional to $\sqrt{V}$. Write a differential equation for $\frac{dV}{dt}$, using $k > 0$.

$\frac{dV}{dt} = -k\sqrt{V}$.

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7 [Foundational] • Target: Forming differential equations from a description

A chemical reaction's concentration C decreases at a rate proportional to the square of the concentration. Write a differential equation for $\frac{dC}{dt}$, using $k > 0$.

$\frac{dC}{dt} = -kC^2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving modeling equations with initial conditions

A population P satisfies $\frac{dP}{dt} = 0.08P$. If $P = 500$ when $t = 0$, find P when $t = 10$, to the nearest whole number.

$P(t) = 500e^{0.08t}$. $P(10) = 500e^{0.8} = 500(2.225541) \approx 1113$.

9 [Intermediate] • Target: Solving modeling equations with initial conditions

Solve $\frac{dV}{dt} = -k\sqrt{V}$ (from the tank problem above), given $V(0) = V_0$.

Separate: $V^{-1/2}\,dV = -k\,dt$. Integrate: $2\sqrt{V} = -kt + C$. Using $V(0) = V_0$: $C = 2\sqrt{V_0}$. So $\sqrt{V} = \sqrt{V_0} - \frac{kt}{2}$, giving $V(t) = \left(\sqrt{V_0} - \frac{kt}{2}\right)^2$.

10 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + 3y = 12$ using an integrating factor, given $y = 2$ when $x = 0$.

Integrating factor: e^{3x}. $\frac{d}{dx}(ye^{3x}) = 12e^{3x}$. Integrate: $ye^{3x} = 4e^{3x} + C$, so $y = 4 + Ce^{-3x}$. Using $y = 2, x = 0$: $C = -2$. Solution: $y = 4 - 2e^{-3x}$.

11 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} - 2y = e^{3x}$ using an integrating factor, given $y = 1$ when $x = 0$.

Integrating factor: e^{-2x}. $\frac{d}{dx}(ye^{-2x}) = e^{3x}e^{-2x} = e^x$. Integrate: $ye^{-2x} = e^x + C$, so $y = e^{3x} + Ce^{2x}$. Using $y = 1, x = 0$: $C = 0$. Solution: $y = e^{3x}$.

12 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + y = 5$ using an integrating factor, given $y = 0$ when $x = 0$.

Integrating factor: e^x. $\frac{d}{dx}(ye^x) = 5e^x$. Integrate: $ye^x = 5e^x + C$, so $y = 5 + Ce^{-x}$. Using $y = 0, x = 0$: $C = -5$. Solution: $y = 5 - 5e^{-x}$.

13 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + 4y = 8$ using an integrating factor, given $y = 3$ when $x = 0$.

Integrating factor: e^{4x}. $\frac{d}{dx}(ye^{4x}) = 8e^{4x}$. Integrate: $ye^{4x} = 2e^{4x} + C$, so $y = 2 + Ce^{-4x}$. Using $y = 3, x = 0$: $C = 1$. Solution: $y = 2 + e^{-4x}$.

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14 [Intermediate] • Target: Further integrating factor examples

Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$ for $x > 0$ using an integrating factor, given $y = 1$ when $x = 1$.

Integrating factor: $e^{\int \frac{1}{x} dx} = x$. $\frac{d}{dx}(xy) = x^3$. Integrate: $xy = \frac{x^4}{4} + C$, so $y = \frac{x^3}{4} + \frac{C}{x}$. Using $y = 1, x = 1$: $C = \frac{3}{4}$. Solution: $y = \frac{x^3}{4} + \frac{3}{4x}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Further integrating factor examples • Cross-Lesson: 12.01 Poisson Distribution

Solve $\frac{dy}{dx} - \frac{2y}{x} = x^3$ for $x > 0$ using an integrating factor, given $y = 3$ when $x = 1$.

Integrating factor: $e^{\int -\frac{2}{x} dx} = x^{-2}$. $\frac{d}{dx}(yx^{-2}) = x$. Integrate: $yx^{-2} = \frac{x^2}{2} + C$, so $y = \frac{x^4}{2} + Cx^2$. Using $y = 3, x = 1$: $C = \frac{5}{2}$. Solution: $y = \frac{x^4}{2} + \frac{5}{2}x^2$.

16 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 13.01 Continuous Random Variables

Verify that $y = 3e^{2x}$ satisfies $\frac{dy}{dx} - 2y = 0$.

$\frac{dy}{dx} = 6e^{2x}$. Substituting: $6e^{2x} - 2(3e^{2x}) = 6e^{2x} - 6e^{2x} = 0$, confirming the equation is satisfied.

17 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 14.01 Sampling & Estimation

Verify that $y = x^2 + Ce^{-x}$ is a general solution of $\frac{dy}{dx} + y = x^2 + 2x$.

$\frac{dy}{dx} = 2x - Ce^{-x}$. Substituting: $(2x - Ce^{-x}) + (x^2 + Ce^{-x}) = 2x + x^2$, which matches the right-hand side $x^2 + 2x$ exactly, confirming the equation is satisfied for any constant C.

18 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 15.01 Hypothesis Testing

Verify that $y = \frac{1}{2}\sin(2x)$ satisfies $\frac{d^2y}{dx^2} + 4y = 0$.

$\frac{dy}{dx} = \cos(2x)$. $\frac{d^2y}{dx^2} = -2\sin(2x)$. Substituting: $-2\sin(2x) + 4\left(\frac{1}{2}\sin(2x)\right) = -2\sin(2x) + 2\sin(2x) = 0$, confirming the equation is satisfied.

19 [Challenging] • Target: General and particular solutions • Cross-Lesson: 2.01 Exponentials & Logarithms

Find the general solution of $\frac{dy}{dx} = 4x^3 - 6x$, and explain what determines a particular solution from this family.

Integrating directly: $y = x^4 - 3x^2 + C$. A particular solution is obtained by specifying one point that the curve must pass through, which fixes the value of the arbitrary constant C.

20 [Challenging] • Target: General and particular solutions • Cross-Lesson: 3.01 Advanced Trigonometry

A curve satisfies $\frac{dy}{dx} = 6x^2 - 4$ and passes through $(1, -1)$. Find the equation of this particular solution.

$y = 2x^3 - 4x + C$. Using $(1, -1)$: $-1 = 2 - 4 + C \Rightarrow C = 1$. Solution: $y = 2x^3 - 4x + 1$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Separation of Variables

Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with initial condition $y(0) = 4$.

a. $y^2 = 2x^2 + 16$ ✓

b. $y^2 = x^2 + 16$

c. $y = 2x + 4$

d. $y^2 = 2x^2 + 4$

Separate variables: $ydy = 2xdx$. Integrate: $\frac{y^2}{2} = x^2 + C \Rightarrow y^2 = 2x^2 + 2C$. At $(0, 4)$: $16 = 0 + 2C \Rightarrow 2C = 16$. Solution: $y^2 = 2x^2 + 16$.

22 [Foundational] • Target: Separation of Variables

Find the general solution of $\frac{dy}{dx} = xy$.

a. $y = Ae^{x^2/2}$ ✓

b. $y = Ae^{x^2}$

c. $y = \frac{1}{2}x^2 + C$

d. $y = Ae^x$

$\frac{1}{y}dy = xdx \Rightarrow \ln|y| = \frac{x^2}{2} + c \Rightarrow y = Ae^{x^2/2}$.

23 [Foundational] • Target: Separation of Variables

Solve the differential equation $\frac{dy}{dx} = 1 + y^2$.

a. $y = \tan(x + C)$ ✓

b. $y = \ln(1 + x) + C$

c. $y = e^x - 1$

d. $y = \arctan(x) + C$

$\frac{1}{1+y^2}dy = dx \Rightarrow \arctan(y) = x + C \Rightarrow y = \tan(x + C)$.

24 [Foundational] • Target: First-Order Equations Requiring Partial Fractions

Find the general solution of $\frac{dy}{dx} = y(1 - y)$.

a. $y = \frac{Ae^x}{1 + Ae^x}$ ✓

b. $y = Ae^x - A$

c. $y = \ln|1 - y| + x$

d. $y = \frac{1}{1 + Ae^{-x}}$

$\frac{1}{y(1-y)}dy = dx \Rightarrow (\frac{1}{y} + \frac{1}{1-y})dy = dx \Rightarrow \ln|y| - \ln|1 - y| = x + c \Rightarrow \ln\left|\frac{y}{1-y}\right| = x + c \Rightarrow \frac{y}{1-y} = Ae^x \Rightarrow y = \frac{Ae^x}{1+Ae^x} = \frac{1}{1+A^{-1}e^{-x}}$.

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25 [Foundational] • Target: First-Order Equations Requiring Partial Fractions

Solve $\frac{dy}{dt} = k(M - y)$ with $y(0) = 0$ for constant $k, M > 0$.

a. $y = M(1 - e^{-kt})$ ✓

b. $y = Me^{-kt}$

c. $y = M - e^{kt}$

d. $y = M(1 + e^{-kt})$

$\frac{1}{M-y}dy = kdt \Rightarrow -\ln|M - y| = kt + c \Rightarrow M - y = Ae^{-kt}$. Since $y(0) = 0$, $A = M$. Thus $y = M(1 - e^{-kt})$.

26 [Foundational] • Target: First-Order Equations Requiring Partial Fractions

A chemical reaction produces a substance according to $\frac{dx}{dt} = 2(4 - x)^2$ with $x(0) = 0$. Find $x(t)$.

a. $x(t) = 4 - \frac{4}{8t+1}$ ✓

b. $x(t) = 4(1 - e^{-2t})$

c. $x(t) = \frac{8t}{2t+1}$

d. $x(t) = 4 - \frac{1}{2t}$

$\frac{1}{(4-x)^2}dx = 2dt \Rightarrow \frac{1}{4-x} = 2t + C$. At $t = 0$: $1/4 = C$. $\frac{1}{4-x} = 2t + \frac{1}{4} = \frac{8t+1}{4} \Rightarrow 4 - x = \frac{4}{8t+1} \Rightarrow x = 4 - \frac{4}{8t+1}$.

27 [Foundational] • Target: Rate of Change Modeling Problems

The rate of decay of a substance is proportional to its remaining mass m. Write down the differential equation governing $m(t)$ for constant $k > 0$.

a. $\frac{dm}{dt} = -km$ ✓

b. $\frac{dm}{dt} = km$

c. $\frac{dm}{dt} = -k$

d. $\frac{d^2m}{dt^2} = -km$

Rate of decay means rate of change is negative and proportional to mass: $\frac{dm}{dt} = -km$.

28 [Foundational] • Target: Rate of Change Modeling Problems

Water leaks from a cylindrical tank of cross-sectional area A through a hole at the bottom according to Torricelli's Law: $\frac{dh}{dt} = -c\sqrt{h}$. If initial height is H, how long does it take to empty completely?

a. $T = \frac{2\sqrt{H}}{c}$ ✓

b. $T = \frac{\sqrt{H}}{c}$

c. $T = \frac{H}{c}$

d. $T = \frac{H^2}{2c}$

$h^{-1/2}dh = -cdt \Rightarrow 2\sqrt{h} = -ct + C$. At $t = 0$, $C = 2\sqrt{H}$. Empty when $h = 0 \Rightarrow 0 = -ct + 2\sqrt{H} \Rightarrow t = \frac{2\sqrt{H}}{c}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Rate of Change Modeling Problems

A population P grows according to the logistic model $\frac{dP}{dt} = 0.04P\left(1 - \frac{P}{2000}\right)$. What is the carrying capacity of the population?

a. **2000** ✓

b. 0.04

c. 1000

d. 50000

The logistic equation is $\frac{dP}{dt} = rP(1 - P/K)$, where K is the carrying capacity. Here $K = 2000$.

30 [Intermediate] • Target: Particular Solutions & Long-Term Behavior

For the logistic model in the previous question, at what population size P is the growth rate $\frac{dP}{dt}$ maximized?

a. $P = 1000$ ✓

b. $P = 2000$

c. $P = 500$

d. $P = 1500$

The growth rate is a quadratic function of P with zeros at 0 and K. Its vertex (maximum) occurs at exactly half the carrying capacity: $P = K/2 = 2000/2 = 1000$.

31 [Intermediate] • Target: Particular Solutions & Long-Term Behavior

Solve the differential equation $x\frac{dy}{dx} = y + 1$ with $y(1) = 3$.

a. $y = 4x - 1$ ✓

b. $y = 3x$

c. $y = 4x^2 - 1$

d. $y = 2x + 1$

$\frac{1}{y+1}dy = \frac{1}{x}dx \Rightarrow \ln|y+1| = \ln|x| + c \Rightarrow y + 1 = Ax$. At $(1, 3)$: $3 + 1 = A(1) \Rightarrow A = 4$. Solution is $y = 4x - 1$.

32 [Intermediate] • Target: Particular Solutions & Long-Term Behavior

A body cools according to $\frac{dT}{dt} = -0.1(T - 20)$. What does $T(t)$ approach as $t \to \infty$?

a. $20\,^\circ\text{C}$ ✓

b. $0\,^\circ\text{C}$

c. ∞

d. $-20\,^\circ\text{C}$

The solution is $T(t) = 20 + Ce^{-0.1t}$. As $t \to \infty$, $e^{-0.1t} \to 0$, so $T(t) \to 20\,^\circ\text{C}$ (ambient temperature).

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33 [Intermediate] • Target: Particular Solutions & Long-Term Behavior

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 23$ and $T_0 = 63$. Find T at the time t when $e^{-kt} = 1/4$.

a. **33** ✓

b. 63

c. 34

d. 13

$T = 23 + (63 - 23) \times \frac{1}{4} = 23 + 40/4 = 23 + 10 = 33$.

34 [Intermediate] • Target: Solving by separation of variables

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 25$ and $T_0 = 41$. Find T at the time t when $e^{-kt} = 1/4$.

a. 41

b. 21

c. **29** ✓

d. 30

$T = 25 + (41 - 25) \times \frac{1}{4} = 25 + 16/4 = 25 + 4 = 29$.

35 [Intermediate] • Target: Forming differential equations from a description

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 25$ and $T_0 = 37$. Find T at the time t when $e^{-kt} = 1/4$.

a. 37

b. **28** ✓

c. 29

d. 22

$T = 25 + (37 - 25) \times \frac{1}{4} = 25 + 12/4 = 25 + 3 = 28$.

36 [Intermediate] • Target: Solving modeling equations with initial conditions

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 17$ and $T_0 = 27$. Find T at the time t when $e^{-kt} = 1/2$.

a. 27

b. **22** ✓

c. 12

d. 23

$T = 17 + (27 - 17) \times \frac{1}{2} = 17 + 10/2 = 17 + 5 = 22$.

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37 [Intermediate] • Target: First-order linear differential equations using an integrating factor

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 19$ and $T_0 = 31$. Find T at the time t when $e^{-kt} = 1/3$.

a. **23** ✓

b. 24

c. 15

d. 31

$T = 19 + (31 - 19) \times \frac{1}{3} = 19 + 12/3 = 19 + 4 = 23$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Further integrating factor examples • Cross-Lesson: 15.01 Hypothesis Testing

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 16$ and $T_0 = 28$. Find T at the time t when $e^{-kt} = 1/2$.

a. 23

b. **22** ✓

c. 28

d. 10

$T = 16 + (28 - 16) \times \frac{1}{2} = 16 + 12/2 = 16 + 6 = 22$.

39 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 2.01 Exponentials & Logarithms

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 18$ and $T_0 = 36$. Find T at the time t when $e^{-kt} = 1/3$.

a. 25

b. 12

c. 36

d. **24** ✓

$T = 18 + (36 - 18) \times \frac{1}{3} = 18 + 18/3 = 18 + 6 = 24$.

40 [Challenging] • Target: General and particular solutions • Cross-Lesson: 3.01 Advanced Trigonometry

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 19$ and $T_0 = 35$. Find T at the time t when $e^{-kt} = 1/4$.

a. 35

b. 24

c. 15

d. **23** ✓

$T = 19 + (35 - 19) \times \frac{1}{4} = 19 + 16/4 = 19 + 4 = 23$.

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41 [Challenging] • Target: Applications: Newton's law of cooling • Cross-Lesson: 4.01 Differentiation

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 13$ and $T_0 = 37$. Find T at the time t when $e^{-kt} = 1/6$.

a. 9

b. 18

c. **17** ✓

d. 37

$T = 13 + (37 - 13) \times \frac{1}{6} = 13 + 24/6 = 13 + 4 = 17$.

42 [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 5.01 Integration

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 24$ and $T_0 = 48$. Find T at the time t when $e^{-kt} = 1/3$.

a. 16

b. **32** ✓

c. 33

d. 48

$T = 24 + (48 - 24) \times \frac{1}{3} = 24 + 24/3 = 24 + 8 = 32$.

43 [Challenging] • Target: Separation of Variables • Cross-Lesson: 6.01 Sequences & Series

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 24$ and $T_0 = 36$. Find T at the time t when $e^{-kt} = 1/6$.

a. 22

b. 36

c. 27

d. **26** ✓

$T = 24 + (36 - 24) \times \frac{1}{6} = 24 + 12/6 = 24 + 2 = 26$.

44 [Challenging] • Target: First-Order Equations Requiring Partial Fractions • Cross-Lesson: 7.01 Vectors

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 23$ and $T_0 = 35$. Find T at the time t when $e^{-kt} = 1/3$.

a. **27** ✓

b. 19

c. 35

d. 28

$T = 23 + (35 - 23) \times \frac{1}{3} = 23 + 12/3 = 23 + 4 = 27$.

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45 [Challenging] • Target: Rate of Change Modeling Problems • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 21$ and $T_0 = 45$. Find T at the time t when $e^{-kt} = 1/3$.

a. 13

b. 30

c. **29** ✓

d. 45

$T = 21 + (45 - 21) \times \frac{1}{3} = 21 + 24/3 = 21 + 8 = 29$.

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